

How Do The Trump Tariffs Affect Carbon Emissions?

Sean Hwang*

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Abstract

This paper examines the impact of the Trump administration's tariffs on global carbon emissions using a global computable general equilibrium framework and decomposes emissions changes into three channels: the aggregate output channel, the sectoral mix channel, and the emissions intensity channel. While all three channels contribute to a reduction in global emissions, the emissions intensity channel accounts for the largest share of the decline, largely through increases in renewable power generation and substitution away from fossil fuels. The results suggest that trade policy can interact with climate policy in unexpected ways, at times accelerating the shift to cleaner production.

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*Brandeis University, Department of Economics. Email: siyounghwang@brandeis.edu

1 Introduction

How do the Trump tariffs affect carbon emissions? This paper answers this question by quantifying and identifying the channels through which recent United States tariff policy influences carbon emissions. Given the urgency of climate change mitigation, understanding how major policy interventions shape emissions is of critical importance. While many studies have examined the effects of the tariffs on economic outcomes like GDP, employment, and welfare, their environmental consequences have received far less attention.

The Trump tariffs represent one of the most significant trade policy shifts in recent history, with real and measurable impacts on global production patterns. At first glance, tariffs might be expected to reduce emissions simply by lowering trade volumes and slowing economic activity. Indeed, my findings show that overall emissions decline. However, the channel through which these reductions occur is not straightforward. On the other hand, trade barriers can also impede the international supply chains for environmental goods and services (EGS), many of which are predominantly produced in China, raising costs and slowing the global transition to cleaner production.

This paper decomposes the tariff-induced changes in emissions into three channels: the aggregate output channel, the sectoral mix channel, and the emissions intensity channel. The aggregate output channel captures emissions reductions associated with contractions in aggregate output caused by tariffs reducing trade and economic activity. The sectoral mix channel reflects changes in the sectoral composition of production within regions. For example, tariff-induced shifts toward cleaner sectors in the United States reduce emissions through this channel, though shifts toward more carbon-intensive sectors could have the opposite effect. Finally, the emissions intensity channel operates through changes in the emissions intensity of production, driven by changes in the use of renewable energy sources and non-energy inputs. This channel illustrates how tariffs may indirectly reinforce existing green policies by raising the share of renewable power generation and encouraging substitution away from fossil fuels toward non-energy factors of production, such as labor and capital. The emissions intensity channel accounts for the majority of emissions changes, while the aggregate output and sectoral mix channels are comparatively small.

An important caveat of this paper is that the results are not a forecast, but rather model-dependent scenario outcomes arising from strong modeling assumptions. One key assumption is that environmental policies remain constant while tariffs change, meaning that policy changes affecting environmental regulation are not explicitly incorporated. This assumption is a strong driver of the model's results. Relaxing this assumption could materially shrink the reduction and could potentially reverse the result to a net increase in emissions.

While this assumption does not fully capture reality, its purpose is to isolate the effect of tariffs on carbon emissions, which is itself an important question to consider. The findings under this assumption highlight the importance of sustained commitment to green policies, as trade interventions can interact with and, in some cases, amplify their effectiveness. The decomposition framework plays a central role in explaining this result. By distinguishing between the three channels, it iden-

tifies the precise channels through which tariffs influence emissions. This methodology moves the analysis beyond documenting aggregate emissions outcomes and uncovers the mechanisms through which trade policy reshapes global environmental outcomes.

2 Background: Trump Tariffs

President Trump has used tariffs as a cornerstone of his trade policy, once remarking that “tariff is the most beautiful word in the dictionary.” On April 2, 2025, the administration enacted the largest tariff increase since the Smoot-Hawley Act of 1930 (Evenett and Fritz, 2025; Clausing, 2025). The tariffs were justified on the grounds that persistent and large U.S. trade deficits posed a national security risk. The executive order implemented a 10 percent ad valorem baseline tariff on all foreign-origin imports, with exceptions for certain goods under the United States–Mexico–Canada Agreement (USMCA). In addition, higher reciprocal tariffs were imposed on several countries to address bilateral trade imbalances, while specific industries such as aluminum, autos, and steel faced a 25 percent tariff.¹ The reciprocal tariff was defined as the tariff rate required to balance bilateral trade between the United States and a trading partner, calculated as

$$\Delta\tau_i = \frac{x_i - m_i}{\epsilon * \phi * m_i} \quad (1)$$

where ϵ is the elasticity of imports with respect to import prices, ϕ is the tariff pass-through rate to import prices, m_i is total imports from country i , and x_i is total exports (Office of the United States Trade Representative, 2025). This is the claim advanced by the USTR, however, many economists contend that it reflects a fundamental misunderstanding of international trade and that the tariffs are “designed for maximum damage to America” (Obstfeld, 2025; PIIE, 2025).

In response to these U.S. tariffs, some trading partners sought exemptions or negotiated new trade deals, while others threatened retaliation. South Korea and Japan reached agreements setting tariff rates at 15 percent. The European Union initially threatened to target selected U.S. exports but ultimately negotiated a trade deal that also reduced tariffs to 15 percent. China, by contrast, responded with retaliatory tariffs that escalated rapidly, reaching rates as high as 150 percent in certain sectors by April 11. The escalation subsided on May 12, when both sides agreed to a 90-day pause that lowered the general tariff rate and was subsequently extended for another 90 days, until November 10 (Bown, 2025; J.P. Morgan Global Research, 2025; Lowell et al., 2025; York and Durante, 2025).

Following the initial announcement on April 2, the administration delayed the implementation of the reciprocal tariffs twice. The tariffs were first paused on April 9 for 90 days and again on July 7, pushing implementation to August 1. While acknowledging that the tariff schedule may still be revised, this paper focuses on the version in effect as of August 1. The full schedule of United States tariffs is reported in Table 1 below.

¹Later increased to 50 percent

Table 1: Additional Reciprocal Tariff on Exports to US

Sector	chn	as5	sel	ind	xsa	cca	dea	ocn	xna	lac	eur	meo	mna	afr	rus	row
Crops	34	19	20	50	20	18	17	11	5	11	15	10	15	15	10	14
Livestock	34	19	21	50	16	10	17	10	2	12	14	10	14	15	10	14
Extraction	29	21	21	21	21	21	21	21	2	21	20	21	21	21	21	21
Chemicals	34	17	20	50	20	22	17	10	4	11	18	10	15	24	10	15
Light_mfg	34	18	20	50	20	11	17	10	15	11	15	10	14	16	10	16
En_int_mfg	33	29	33	37	28	44	32	36	19	32	28	31	32	34	39	44
Electrical	34	15	20	50	19	12	17	10	8	12	17	10	15	22	10	15
Machinery	32	16	20	50	19	10	16	10	7	12	16	10	15	25	10	16
Trans_eq	38	22	22	29	24	24	24	15	17	15	16	18	20	25	10	23
TnD	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Construction	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Services	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Dwellings	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Oth_tp	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Wat_tp	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Air_tp	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Coal	34	19	20	50	15	16	19	10	0	10	15	10	15	19	10	15
Oil	34	19	21	50	10	13	15	13	0	12	14	19	29	15	10	12
Gas	10	17	40	50	10	13	18	10	0	15	14	10	26	15	10	15
Oil_Pcts	34	16	20	50	18	23	16	10	0	12	14	13	29	19	10	11
NuclearE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
CoalE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
GasE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WindE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
HydroE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OilE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OthE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SolarE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Source: Trade Compliance Resource Hub. (2025, October 10). Trump 2.0 tariff tracker.

<https://www.tradecomplianceresourcehub.com/2025/10/10/trump-2-0-tariff-tracker>

Additional tariff rates are computed as trade-weighted averages of applied U.S. tariff rates, using import values from each region as weights.

The tariff schedule highlights important regional and sectoral asymmetries. China and India face relatively higher tariffs than most other regions, with India subject to the highest rate at 50 percent because of its purchases of Russian oil and weapons. By contrast, Canada and Mexico (grouped under North America, nec) experience comparatively favorable treatment due to preferential access under USMCA. At the sectoral level, tariffs are concentrated in energy-intensive manufacturing such as steel and aluminum production, as well as transportation equipment, which includes automobiles. Sizable tariffs are also imposed on fossil fuels such as coal, oil, gas, and oil products, raising their prices relative to renewable energy sources and electricity generation. Electricity generation, however, is largely non-tradable across borders and therefore not subject to tariffs, while other sectors such as services, construction, dwellings, and many transport subsectors are similarly exempt. Taken together, these factors contribute to the results by making carbon-intensive energy sources relatively more expensive than less carbon-intensive alternatives, leading

firms to substitute toward cleaner energy sources, as discussed in the results section below. To keep the analysis tractable, the discussion that follows presents detailed results for the World, the United States, India, and China which together provide representative cases for the broader set of regions.

3 Literature Review

Several papers have examined the economic and environmental consequences of the Trump tariffs. McKibbin et al. (2025) find that the tariffs depreciate the U.S. dollar, slow both U.S. and global growth, raise inflation, and reduce long-run U.S. employment. At the firm level, Amiti et al. (2019) document higher prices for intermediates and final goods, supply chain disruptions, and reduced imported varieties from the 2018 trade war with China. For households, tariffs place a disproportionate burden on lower and middle-income households and lead to large income losses despite higher tariff revenues (Clausing and Lovely, 2024; Boller et al., 2025; Mahoney and Cummings, 2025; York and Durante, 2025). In terms of environmental consequences, several studies emphasize that tariffs disrupt supply chains for clean technologies such as solar cells, electric vehicles, and wind turbines, raising costs, slowing adoption, and straining international climate cooperation (Sekin, 2025; Brief, 2025). Others suggest that tariffs could redirect trade, with China seeking new markets for its clean energy technology and the EU gaining a competitive edge in the energy transition, while at the same time creating pressures that may lead countries to relax environmental standards to preserve competitiveness (Cherniwchan, 2017). More optimistically, analysts note that higher costs for U.S. energy-intensive industries could hamper fossil-fuel expansion and much like the shutdown during the COVID-19 pandemic, tariffs may benefit the environment by dampening economic activity (Ali, 2025).

As the literature shows, tariffs can affect the environment through multiple channels, and the overall impact depends on the relative magnitudes of these forces. Antweiler et al. (2001) develop a theoretical model showing that freer trade can reduce pollution overall, as scale effects that increase emissions are offset by technique effects that encourage cleaner production. Copeland and Taylor (1994) expand this discussion by highlighting the complex linkages between income, pollution, and trade. They demonstrate that while free trade can increase global pollution, its effects depend on which region growth occurs. Moreover, when environmental policy is set optimally, the pollution generated by higher economic activity can be offset by shifts toward cleaner production methods. Grossman and Krueger (1993) provide empirical evidence on the relative importance of scale, composition, and technique effects, using a CGE model to assess NAFTA’s impact on pollution in Mexico. Their results show that pollution declines with GDP at higher income levels, consistent with the environmental Kuznets curve. While this paper addresses topics similar to those in the existing literature, it develops a methodological approach that is distinct from, though closely related to, the scale–composition–technique framework and uses a CGE model to analyze the general equilibrium effects of the Trump tariffs on both the economy and the environment. The

proposed methodology decomposes emissions changes into three channels—the aggregate output channel, the sectoral mix channel, and the emissions intensity channel—to empirically assess the contribution of each channel. This approach not only documents the outcomes but also uncovers the mechanisms that generate them.

Other studies have focused on the use of tariffs as a policy tool for environmental objectives, particularly to prevent pollution havens and carbon leakage. The pollution haven hypothesis suggests that firms may relocate production to countries with weaker environmental regulations to lower costs, thereby increasing global emissions. However, empirical evidence provides little support for this hypothesis, and tariffs or other trade barriers are generally seen as inefficient environmental policy instruments that represent, at best, second-best solutions (Dean et al., 2009; Cherniwchan et al., 2017; World Bank, 2020b; Levinson, 2023). Carbon leakage refers to a related problem in which stringent climate policies in one country reduce emissions domestically but lead to offsetting increases abroad as energy-intensive production shifts to less regulated regions. A prominent example of tariff-based mitigation is the European Union’s Carbon Border Adjustment Mechanism (CBAM), which seeks to level the playing field by imposing charges on imports based on their carbon content, thereby discouraging offshoring of emission-intensive production (Zhong and Pei, 2024). There is, however, no clear evidence that the Trump tariffs were designed to achieve such environmental policy objectives.

4 Methodology/Model

This section outlines the computable general equilibrium (CGE) model used in the analysis and describes its key components, including the model closure, scenario design, and data. Particular attention is given to the assumptions and mechanisms in the energy sector that are most relevant for interpreting the emissions results.

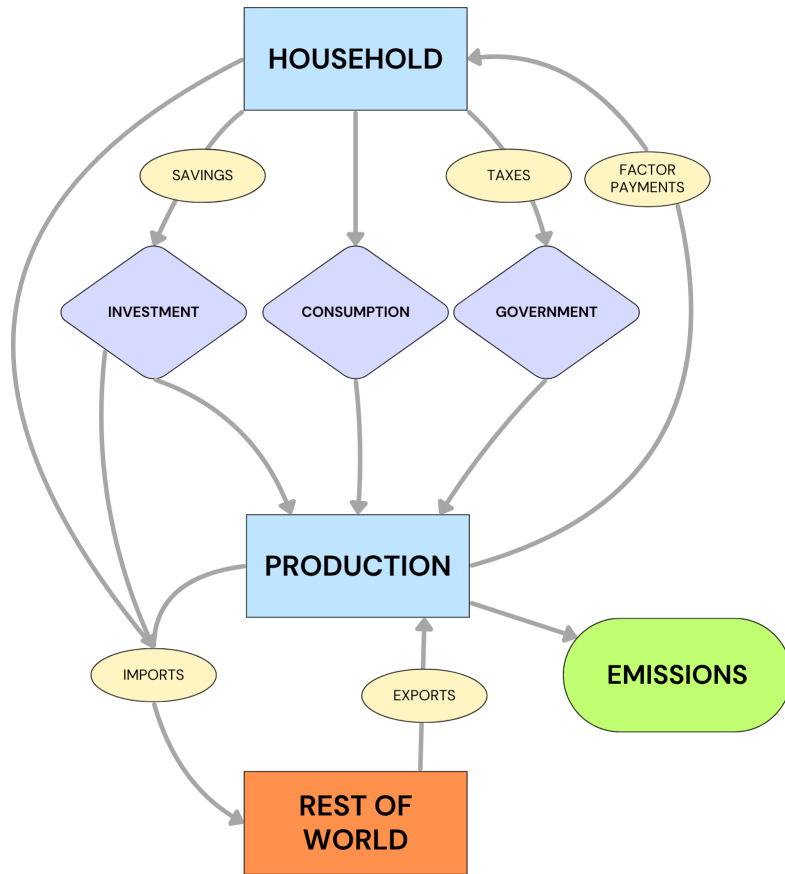
Global Trade-Environment Model

The Global Trade-Environment Model is a dynamic, recursive computable general equilibrium (CGE) model extended with environmental modules, including energy use and emissions accounting. Like other large CGE models, the model used in this paper is composed of over 10,000 simultaneous equations and is described in rigorous algebraic detail in Aguiar et al. (2019). However, given the complexity of providing a full and detailed explanation of the model, this section offers only a concise intuitive overview, focusing on the model components most directly relevant to the environmental simulations.

This model is particularly well suited to analyze the effects of tariffs on emissions because it models the economy in its entirety, representing the interactions between households, firms, and governments within global markets. This framework enables the study of cross-border trade flows and emissions linkages between regions. The model consists of a system of equations derived from economic theory that are summarized schematically in Figure 1. The arrows represent specific types

of equations. For example, the arrows entering the production block denote commodity market demand, defined as the sum of intermediate demand, investment, consumption, and government use. These must equal total output of a good including exports, shown by the arrows leaving the production block. The equilibrium solution ensures that all markets clear simultaneously, with prices adjusting so that supply equals demand across the entire economy. International trade links production with final demand by making imported and exported goods available alongside domestic goods. Tariffs and other trade measures enter as ad valorem wedges between domestic and imported prices, altering relative costs and trade flows.

Figure 1: Circular flow of income and expenditure in the CGE model



Households

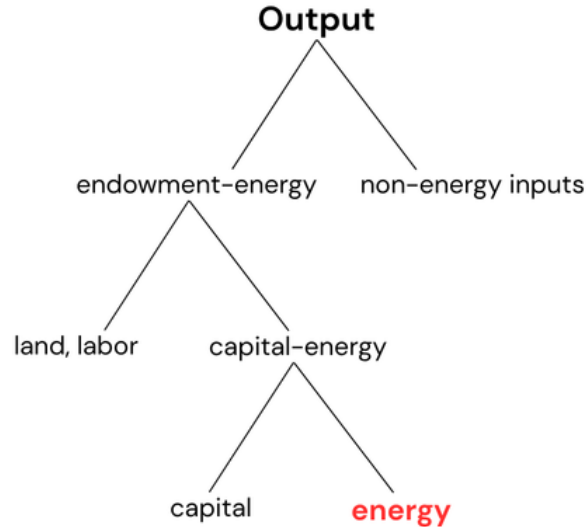
Households receive income from factor payments (wages, returns to capital, and land rents) and allocate it across private consumption, investment in the form of savings, and government spending through taxes (Figure 1). This allocation is governed by a Cobb–Douglas utility function, meaning that a constant share of income is directed to each type of final demand. For savings, the model uses a simple constant marginal consumption–income ratio for each region to determine savings in any given year. This marginal rate changes over time. Investment, on the other hand, is determined by rates of return. Any gap between savings and investment is absorbed by changes in net exports,

which is discussed further in the model closure section. Within private consumption, households allocate expenditures across goods using a non-homothetic Constant Difference Elasticity (CDE) utility function (Aguiar et al., 2019), allowing for varying income elasticities across goods. This structure captures important demand-side dynamics such as the tendency for wealthier households to shift toward cleaner or higher-quality consumption bundles.

Firms

Firms produce goods and services to meet household, government, and investment demand. Firms are assumed to be perfectly competitive, producing homogeneous commodities by combining value-added factors (labor, capital-energy, land, and natural resources) with intermediate inputs. Employment of primary factors depends on the endowments available to each region, which are fixed in each period but evolve over time along exogenously specified paths based on projections from the IMF and other sources. Firms maximize efficiency by minimizing costs given their production technologies. Intermediate inputs are separated into domestic and imported varieties, combined through CES functions under the Armington assumption. This structure ensures that firms can substitute between sources of supply while recognizing that domestic and foreign goods are imperfect substitutes. Domestic production functions are modeled using nested CES structures, which allow substitution between primary factors (land, labor, and capital-energy) and domestic and foreign intermediate inputs (Figure 2).

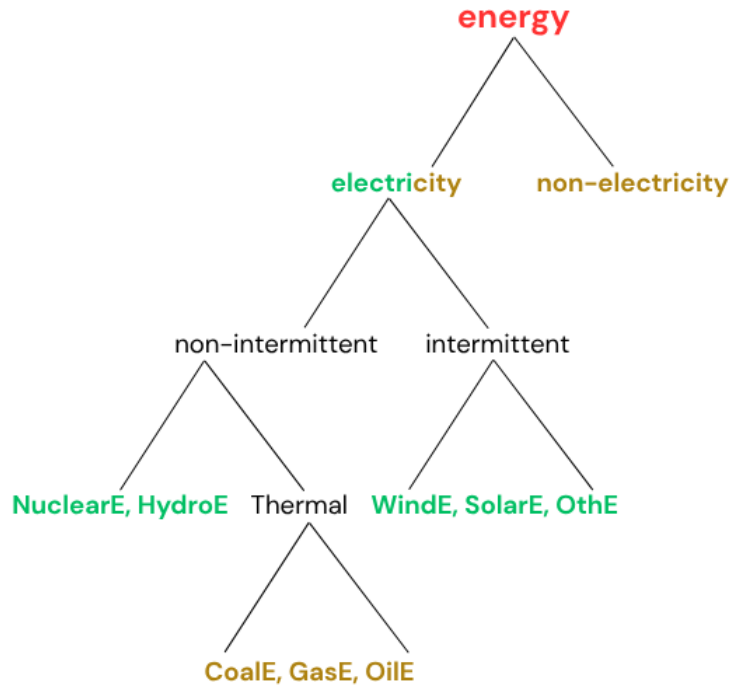
Figure 2: Substitution among inputs in Production



The capital–energy composite is modeled as a CES aggregate, allowing substitution between capital and various forms of energy in response to relative price changes. The energy component is further disaggregated to provide a detailed representation of energy demand, capturing substitution between fossil fuels and renewable sources and their implications for emissions. Composite energy demand is represented as a CES aggregate of electricity and non-electric energy. The non-electric

bundle is further decomposed into coal and non-coal fuels, with the latter split among oil, gas, and refined oil products. Electricity, in particular, is modeled as a mix of fossil-based and renewable sources (Figure 3). Renewables such as wind, solar, and hydro enter the energy bundle alongside fossil fuels, with substitution possibilities governed by CES functions and calibrated elasticities.

Figure 3: Energy Production Nest



CO_2 emissions are produced from both non-electricity and electricity generation. Emissions are calculated from the amount of fossil fuels (coal, oil, gas, and refined oil products) burned directly in both production and consumption activities as well as amounts burned in the different types of electricity generation. The emissions intensity is assumed to be constant for each energy input, though it differs across energy inputs. As a result, total emissions depend on both the composition of energy inputs and the quantity of energy input.

Policies such as tariffs, carbon taxes, and renewable energy subsidies influence relative prices within the nested structure. When the relative price of renewables falls through technological improvements or subsidies, firms substitute away from fossil fuels toward cleaner sources. Likewise, tariffs on imported coal or oil raise their relative prices, encouraging a shift toward domestic energy and renewables. This is one key mechanism through which trade, technology, and policy shocks affect energy demand and emissions outcomes. More details on the theoretical foundations and applications of this model can be found in Aguiar et al. (2019) and Petri et al. (2025).

Model Closure

Given that CGE models are a system of simultaneous equations, the number of independent equations must be equal to the number of endogenous variables. The model closure specifies which variables are determined exogenously outside the model and which are solved endogenously within the system to achieve equilibrium. Some equations explicitly solve for these endogenous variables, while others determine them indirectly. For example, an equation requiring the supply of steel to equal the demand for steel determines the equilibrium price of steel when the market-clearing condition is satisfied. There are three main closures that are important for this model: macroeconomic, employment, and environmental.

The macroeconomic closure determines consumption (C), savings (S), government spending (G), and investment (I). The model includes equations to determine each of these key macroeconomic variables. Consumption, savings, and government spending are determined as functions of total income (Y). Investment is determined by one of four alternative closure specifications described in (Aguilar et al., 2019).

The investment closure used determines investment in each region as a function of the expected regional return to capital. The expected return is positively related to the current return and negatively related to current investment. The model then requires that changes in regional expected returns to equalize to a common global value. This value is determined implicitly as the return that equates global expenditures and global incomes or, equivalently, global savings and investments ².

Regional investments and savings do not necessarily equal, and capital infows (also called foreign savings) into r are defined as $F_r = I_r - S_r$. Since the balance of payments requires that capital flows and the trade balances sum to zero, $I_r - S_r + X_r - M_r = 0$, a regions' trade surplus will equal the negative of its capital inflows, $X_r - M_r = -F_r$. To illustrate, suppose there is a shock that increases the demand for capital in a region and hence raises its rate of return on capital. This will raise the region's expected return above other regions, generate capital inflows, raise investment, and produce a higher capital stock next year. Meanwhile, the country's trade balance will become less positive and the shares of capital and employment allocated to traded products will decline.

For employment, the closure sets the sum of sectoral labor demands equal to a fixed total labor supply, implying full employment, with wages adjusting endogenously to clear the labor market. The choice of employment closure matters because the emissions results depend on the magnitude of the output contraction caused by the tariff. This closure can be altered so that labor supply depends on the real wage rate, making the labor market more rigid. Such rigidities would amplify contractions, leading to larger emissions reductions through the output channel, while also strengthening the emissions intensity channel by further reducing output through labor market adjustments.

For environmental emissions, regional emissions are endogenous. Under this closure, a carbon tax is set exogenously and determines CO_2 emissions through the model's emissions-generation

²In practice the model omits this equation. It is satisfied by Walras' Law, which states that if all but one market are in equilibrium, the remaining market will be in equilibrium as well.

mechanism. This operates by changing the prices and emissions of energy products, such as electricity generation, which in turn affects the emissions generated in the production of goods and services. However, because carbon taxes are set exogenously, the price of carbon does not adjust in response to the tariff. Normally, a decline in income due to the tariff would reduce demand for environmental goods, lowering the carbon price and increasing the relative use of carbon-intensive energy. In this model, however, the fixed carbon price prevents this adjustment, keeping carbon-intensive energy relatively more expensive than renewable energy, especially as renewable energy prices continue to fall due to subsidies. As discussed later, this assumption is a key driver of the results.

Scenarios

The analysis compares two scenarios. The baseline scenario is the IEA’s Stated Policies Scenario (STEPS), which incorporates environmental policies that governments have already enacted but excludes future commitments that have not yet been implemented. The STEPS is not constructed to achieve a particular outcome such as net zero emissions, but rather to trace the environmental implications of prevailing policies and measures as they stand. The policies assessed span a broad spectrum, including Nationally Determined Contributions (NDCs) under the Paris Agreement as well as a wide range of energy-related measures such as pricing regimes, efficiency standards, electrification programs, and infrastructure projects. In the model these policies are implemented as gradual improvements in energy efficiency and electrification, as well as taxes and subsidies that encourage the expansion of wind and solar power while reducing reliance on coal. Further details of the STEPS scenario are available in IEA (2024) and Petri et al. (2025).

The policy scenario introduces the Trump tariffs, implemented on August 1, as a shock following the tariff schedule reported in Table 1. All other assumptions remain the same as in the STEPS baseline. Because environmental policies are fixed by the model closure, they remain unchanged across both scenarios. As a result, this study does not incorporate changes to environmental policy enacted under President Trump. While this assumption does not fully reflect reality, it allows the analysis to focus squarely on the impact of the tariff, leaving a fuller treatment of environmental policy changes to future work. If environmental policies were endogenous, the model would likely predict smaller emissions reductions and possibly a net increase in emissions as a consequence of the tariffs. Even so, the results demonstrate that under the tariff scenario, environmental policy plays a critical role in driving the emissions intensity channel, underscoring the importance of having strong, if not optimal, environmental policies in place.

Data

The model uses the latest dataset from the Global Trade Analysis Project (GTAP-11-POWER-E) aggregated into 17 regions and 28 sectors, of which 13 are energy-related (See Tables 7 and 8 in the Appendix). This level of aggregation allows for a tractable yet detailed representation of international trade and energy use, while capturing the heterogeneity across regions and sectors. The

data include input–output tables that describe inter-industry linkages and form the basis for modeling production, consumption, and trade. They also contain exogenous variables and parameters that govern how variables interact, such as elasticities that determine substitution between inputs and across regions. Energy data from the International Energy Agency (IEA) underpins the STEPS baseline, providing projections of energy use and associated emissions, including improvements in energy efficiency and electrification over time.

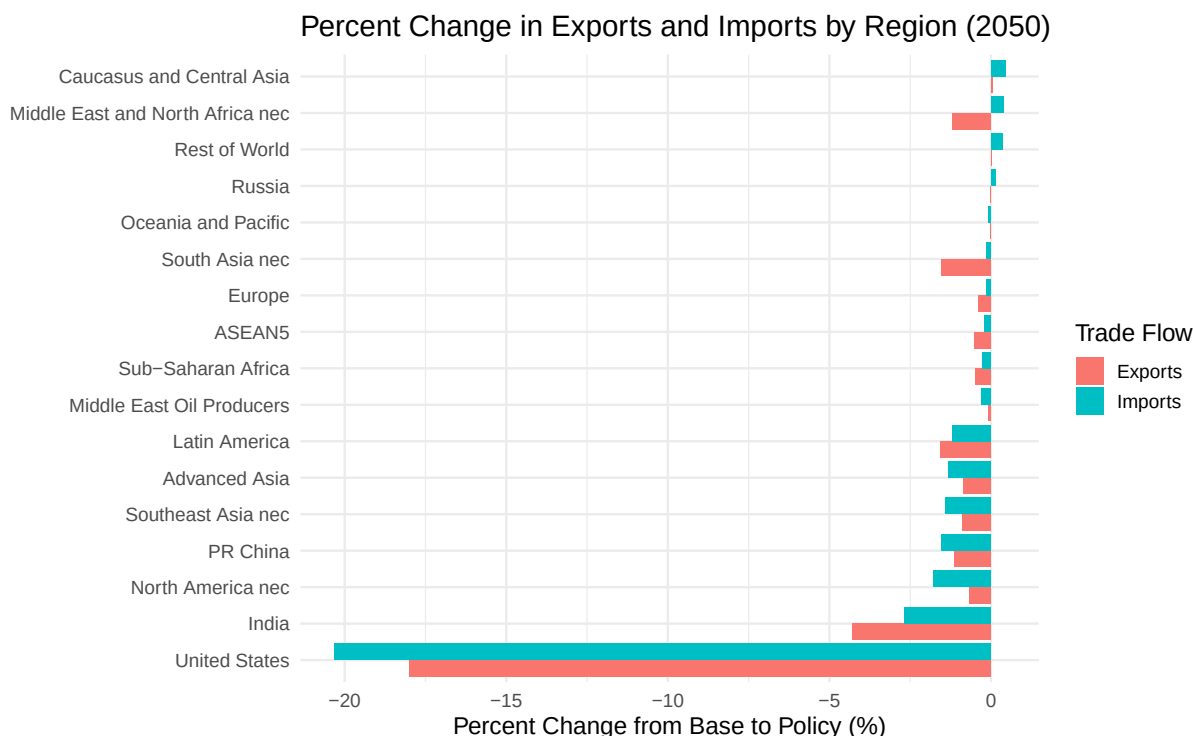
5 Simulation Results

In this section, I report the simulation results of the model, comparing outcomes for trade, GDP, and emissions under the tariff scenario relative to the baseline. All results are reported for 2050 and reflect a simultaneous equilibrium in which all endogenous variables are jointly determined and interact with one another.

Trade

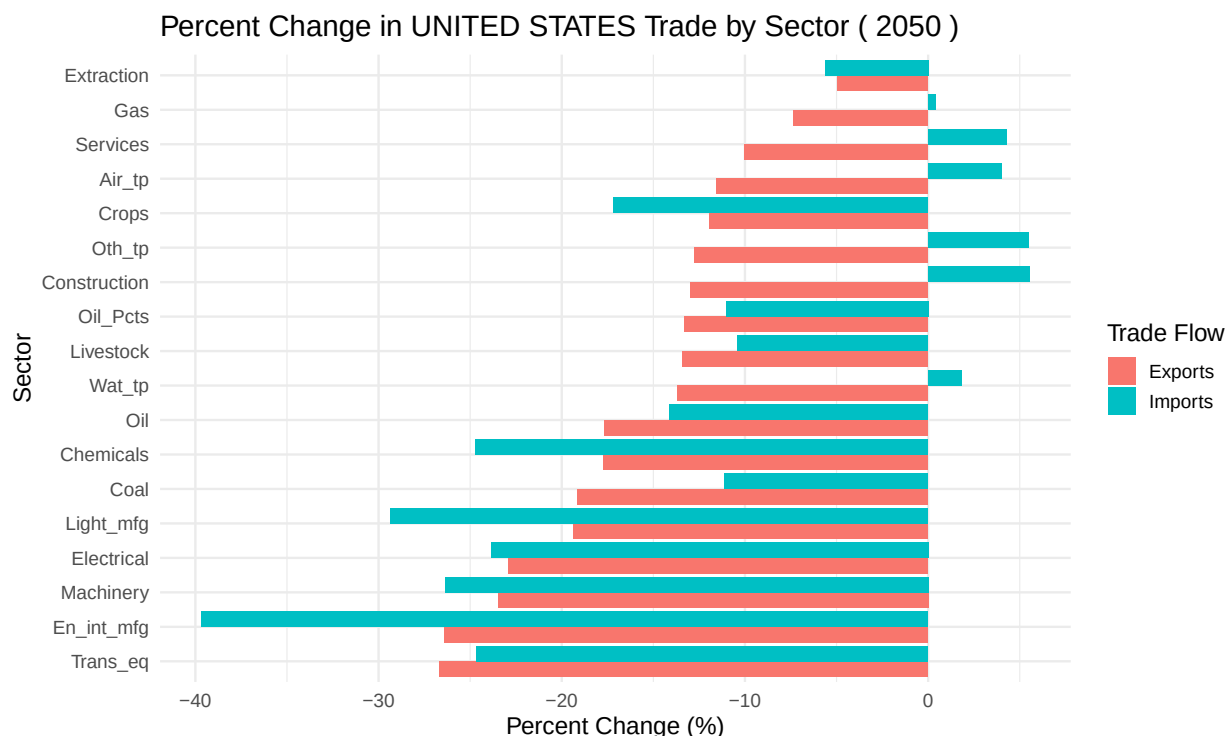
Tariffs significantly shift trade patterns across several regions. Figure 4 shows the percent change in exports and imports across regions in 2050 under the tariff scenario relative to the baseline. The United States is by far the most negatively affected with imports declining more than 20 percent, reflecting the direct impact of tariffs on foreign goods. U.S. exports fall by almost as much, around 17 percent, as global demand for U.S. goods contracts. Notably, no clear beneficiary emerges from the tariffs, suggesting that the costs of tariffs are spread across the global economy. While trade diversion allows some regions to expand exports to new destinations, in the aggregate most regions experience declines in both imports and exports as a result of the tariffs. India is the second most affected economy, facing the highest tariff rate of 50 percent. Its imports fall by 3 percent and exports by 4 percent. Other regions also experience more moderate declines in the range of 1–3 percent. The tariffs exert a global efficiency loss, imposing costs on all major economies, with the heaviest burden falling on the United States. This outcome reflects the general equilibrium nature of tariffs. Although tariffs are imposed to protect domestic industries, they simultaneously distort both imports and exports, generating broad-based efficiency losses not only in the countries directly affected by the tariffs but also across the world economy.

Figure 4: Percent change in exports and imports by region in 2050 under tariff scenario (relative to baseline).



An examination of U.S. trade flows at the sectoral level reveals that the industries most directly targeted by tariffs experience the sharpest declines in both imports and exports (Figure 5). In contrast, sectors not directly affected by tariffs exhibit modest increases in imports, yet their exports continue to contract. The sharpest decline occurs in energy-intensive manufacturing, the sector subject to the highest tariffs, where imports contract by nearly 40 percent. Other heavily targeted industries, including light manufacturing, machinery, chemicals, transport equipment (automobiles), and electrical equipment, also experience substantial import reductions in the range of 25–30 percent. Notably, U.S. exports in many of these same sectors decline by a roughly proportional magnitude, underscoring the broader impact of tariffs on both sides of trade flows.

Figure 5: Percent change in United States exports and imports by sector in 2050 under tariff scenario (relative to baseline).

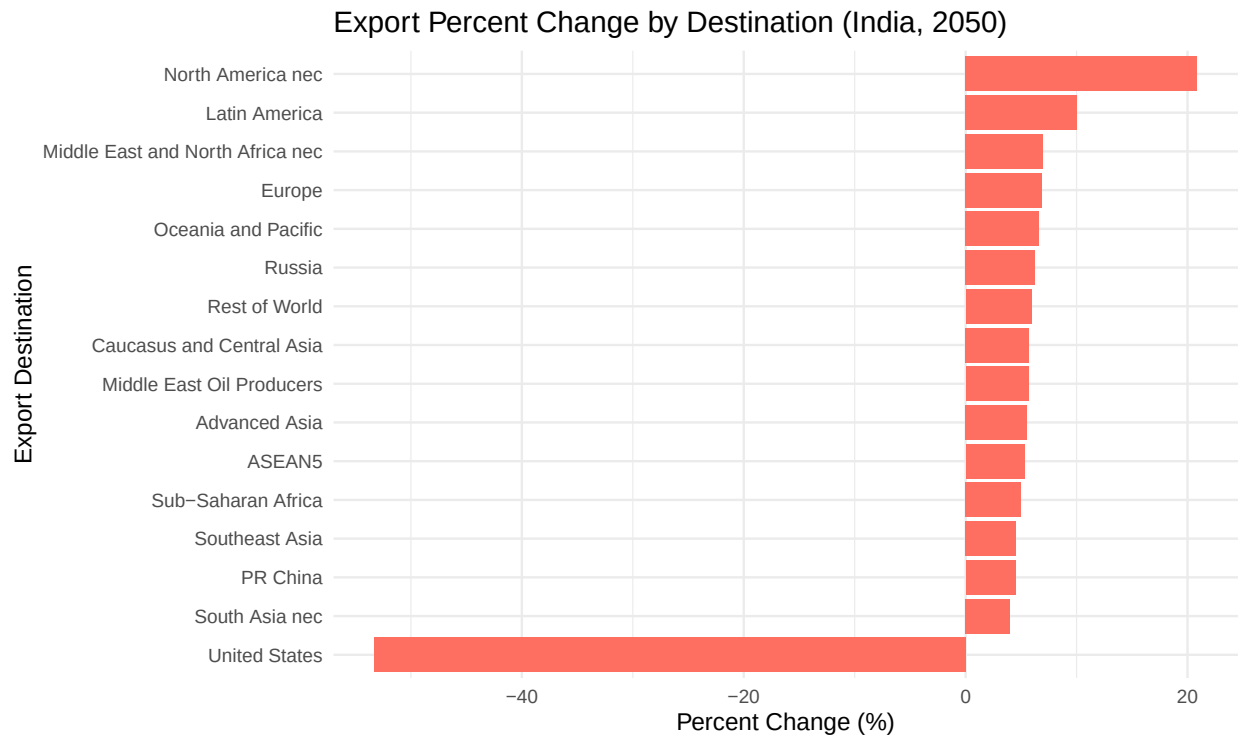


Several mechanisms account for this pattern. First, tariffs raise the cost of imported intermediate inputs, forcing firms to substitute toward more expensive domestic inputs. This increases production costs not only for goods consumed domestically but also for those produced for export, thereby reducing U.S. competitiveness abroad. Second, domestic inputs that might otherwise have been available for export are redirected to meet stronger domestic demand. Third, U.S. trading partners experience income losses as their own exports to the U.S. decline, diminishing their capacity to import U.S. products. Finally, global supply chains are disrupted, raising costs across production networks and further weakening U.S. exports. This is illustrated most clearly by the U.S. service sector. Although not directly targeted by tariffs, services exports, one of the largest and most competitive U.S. sectors, decline, underscoring how broad-based tariffs indirectly erode U.S. export competitiveness through higher costs, disrupted supply chains, and reduced foreign demand.

Figure 6 shows that India's exports to the U.S. in 2050 decline by more than 45 percent. In this section, I focus on India, as it was the region most heavily impacted by the tariffs at the time. Some of India's exports are diverted to North America nec, which benefits from favorable terms of trade with the U.S. relative to other regions. However, this diversion is modest and insufficient to offset the sharp reduction in exports to the United States, even when accounting for increased exports to other regions in the long run. This illustrates the efficiency loss associated with tariffs, where reductions in trade with a major partner are not fully offset elsewhere, resulting in a net

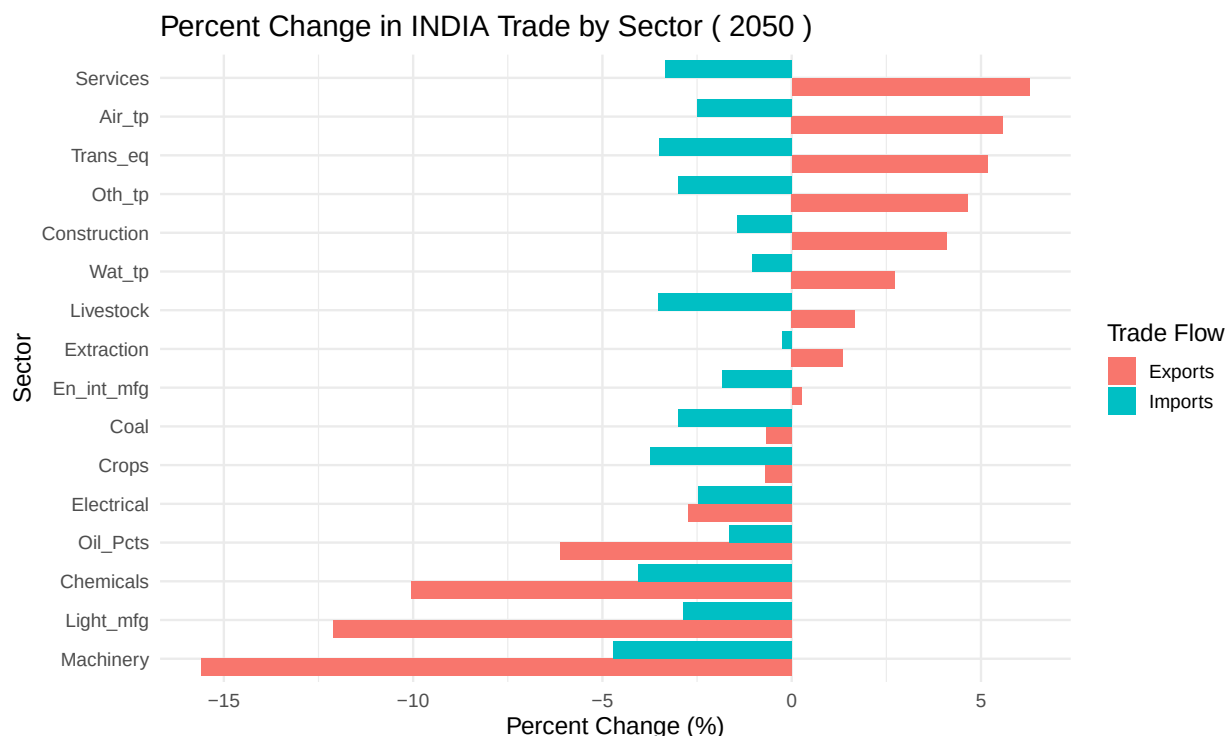
decline in overall trade. This finding highlights that tariffs not only disrupt direct bilateral trade flows but also distort global trade patterns in ways that reduce efficiency and undermine potential gains from trade.

Figure 6: Percent change in India’s exports by destination in 2050 under tariff scenario (relative to baseline).



India’s main export sectors to the United States, machinery, light manufacturing, chemicals, and oil products experience sharp declines in exports, reflecting the direct impact of tariffs (Figure 7). Imports in these same sectors also fall, as reduced production lowers the demand for imported intermediate inputs. This pattern is consistent with the results in the United States, underscoring how global supply chains contract when key sectors are subject to tariffs. At the same time, India experiences increases in exports in sectors not directly targeted. In particular, services, which represent one of India’s largest and most competitive export sectors, show relatively strong growth as resources are reallocated away from manufacturing industries affected by the tariff. Imports into these sectors decline as well, reflecting the redirection of domestic resources toward service production. This reallocation mirrors the sectoral adjustments observed in other regions, such as China, where domestic resources shift into non-tariffed sectors as part of the broader global restructuring induced by tariffs.

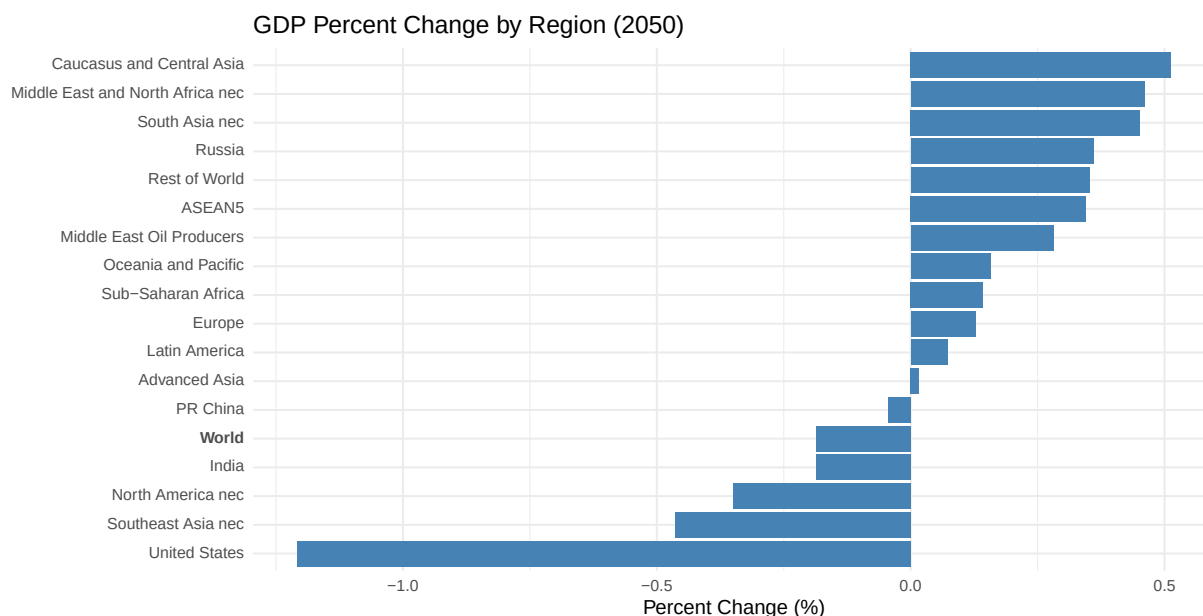
Figure 7: Percent change in India's exports and imports by sector in 2050 under tariff scenario (relative to baseline).



Gross Domestic Product

While tariffs generate substantial changes in trade patterns, their overall effects on GDP are comparatively modest. This reflects the fact that trade accounts for only a small share of GDP, so even large shifts in trade flows translate into relatively small changes in aggregate output. The United States experiences the largest GDP loss, at approximately 1.2 percent, as would be expected given its central role in imposing the tariffs (Figure 8). Other economies also experience GDP declines, including China, India, North America nec, and Southeast Asia nec, reflecting both reduced exports and higher production costs. At the same time, several regions register GDP gains. Europe, ASEAN5, and the Middle East and North Africa benefit slightly as production that was previously concentrated in the United States is diverted to these regions. This reallocation illustrates a general equilibrium outcome. While tariffs generate efficiency losses globally, some regions experience modest gains from the shifting patterns of production and trade.

Figure 8: Percent change in GDP by country and region in 2050 under tariff scenario (relative to baseline).



Emissions

Consistent with the GDP results, changes in emissions under the tariff scenario are small, but not insignificant. Net global emissions decline, with the largest reduction occurring in the United States (Figure 9). In absolute terms, China remains the world's largest emitter, reflecting its scale of production and reliance on fossil fuels. Nevertheless, China experiences an emissions reduction of about 1.4 percent, while North America nec and Europe see declines of roughly 1.9 percent and 1.5 percent, respectively.

Figure 9: Percent change in emissions by country and region in 2050 under tariff scenario (relative to baseline).

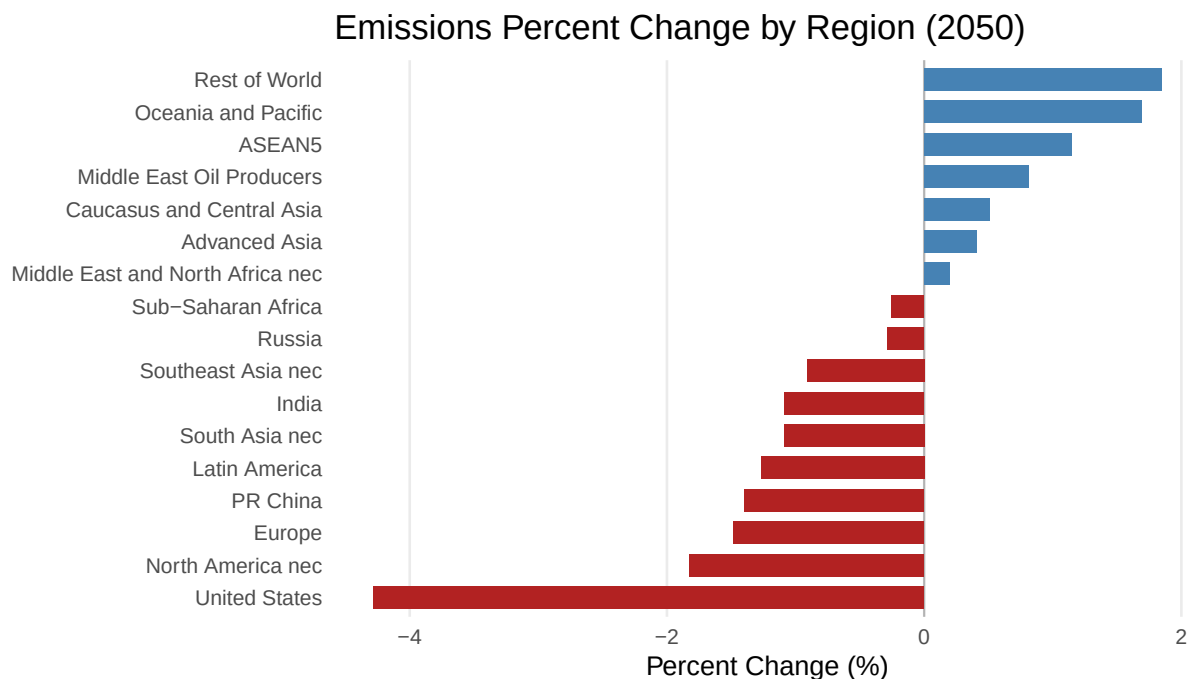
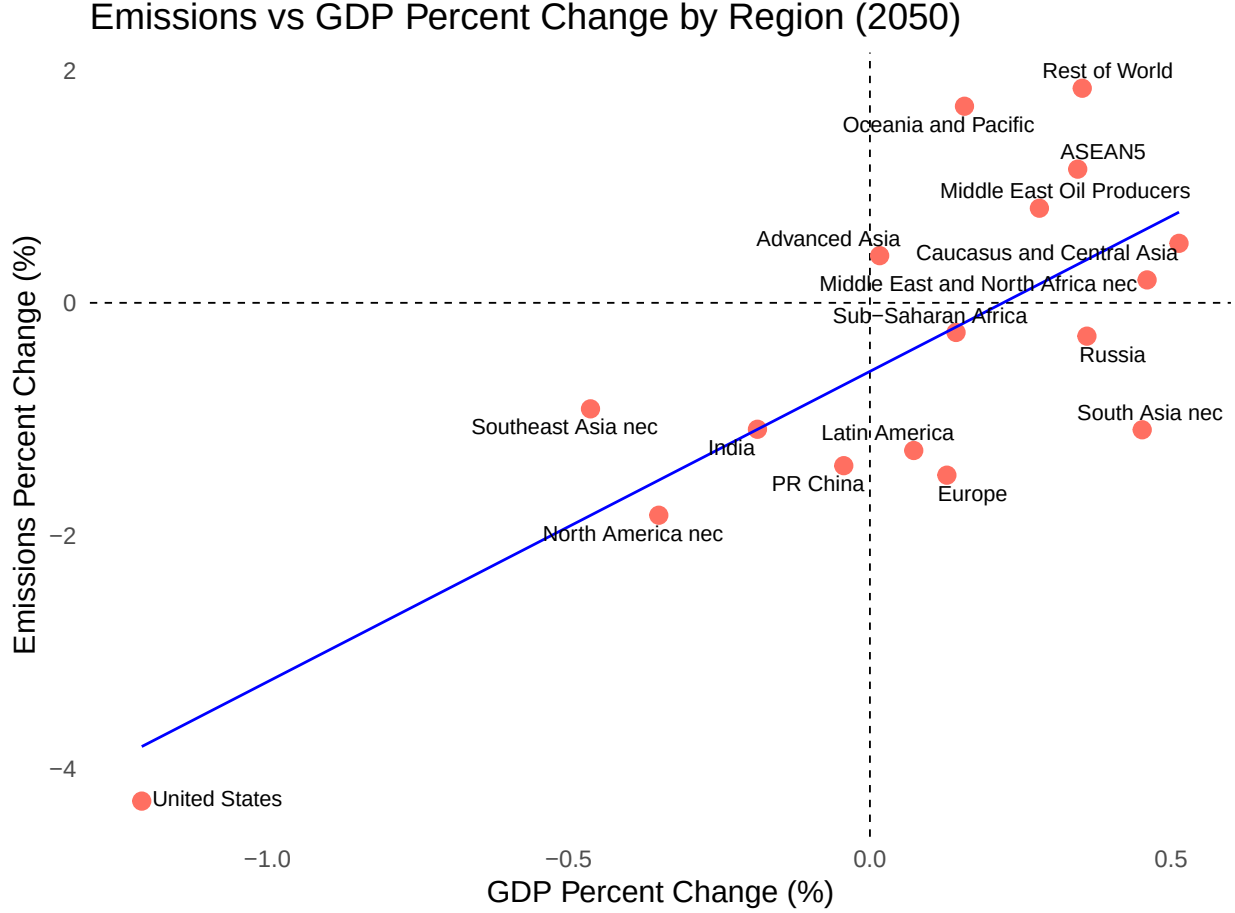


Figure 10 plots the percent change in emissions against the percent change in GDP across regions in 2050 under the tariff scenario. The general pattern aligns with expectations. Regions experiencing GDP growth also tend to register emissions increases, while those with GDP losses exhibit emissions reductions. This outcome reflects the scale effect, whereby higher levels of production are associated with greater energy use and emissions, and conversely, reduced economic activity leads to lower emissions.

Figure 10: Relationship between GDP and emissions changes by region in 2050 under tariff scenario (relative to baseline).



While the scale effect accounts for much of the observed variation, it does not capture the entire picture. Several regions diverge from the one-to-one relationship between GDP and emissions, indicating that additional forces are at work. Therefore, it is necessary to decompose emission changes into the three channels, the aggregate output channel, the sectoral mix channel, and the emissions intensity channel to uncover the mechanisms through which tariffs affect carbon emissions. This decomposition helps isolate the extent to which tariff-induced emission changes are driven by changes in total output, sectoral shifts in production, or adjustments in the emissions intensity of production.

6 Framework

This section presents the framework used to decompose global emissions into three components and discusses how each component is calculated. The framework expresses q_{ij} as the product $e_{ij}s_{ij}r_jG$, and applies a linear approximation to decompose the contribution of each component. Global emissions are the sum of emissions across all sectors and regions (Equation 2). Emissions

from a sector in a given region can be expressed as the product of that sector's emissions intensity, its share of regional output, and the region's share of world GDP. This decomposition provides the basis for decomposing tariff-induced changes in emissions into scale, composition, and technique effects.

$$Q = \sum_i \sum_j q_{ij} = \sum_i \sum_j e_{ij} s_{ij} r_j G \quad (2)$$

Q = Global emissions

q_{ij} = Emissions of sector i , region j

y_{ij} = Output of sector i , region j

$e_{ij} = \frac{q_{ij}}{y_{ij}}$ = Emissions coefficient of sector i , region j

$s_{ij} = \frac{y_{ij}}{g_j}$ = Sector shares (of value added) of region j where $g_j = \sum_i y_{ij}$ (GDP of region j)

$r_j = \frac{g_j}{G}$ = Region shares of World GDP

G = Global GDP

Equation 3 expresses the change in global emissions as the sum of changes in each of the three components weighted by the emissions of sector i in region j . The aggregate output component ($\frac{\Delta r_j}{r_j} + \frac{\Delta G}{G}$) measures the impact on emissions arising from changes in the overall size of the world economy as well as shifts in the regional shares of global GDP. Increased global economic activity raises global emissions. In addition, growth in a region's GDP relative to other regions increases its contribution to global emissions. For example, if China's share of world GDP expands, it will contribute more to global emissions. The sectoral mix component ($\frac{\Delta s_{ij}}{s_{ij}}$) captures how changes in sectors within regions affect emissions. A shift toward carbon-intensive sectors, such as energy-intensive manufacturing, raises emissions, while a shift toward cleaner sectors, such as electrical equipment, lowers them. Finally, the emissions intensity component ($\frac{\Delta e_{ij}}{e_{ij}}$) reflects changes in emissions per unit of output. For example, when firms substitute renewable energy in place of fossil fuels, emissions fall despite output remaining unchanged. Finally, δ denotes the sum of second- and higher-order terms in the linear approximation and is found to be small.

$$\Delta Q = \sum_i \sum_j \left[\underbrace{\frac{\Delta e_{ij}}{e_{ij}}}_{\text{emissions intensity}} + \underbrace{\frac{\Delta s_{ij}}{s_{ij}}}_{\text{sectoral mix}} + \underbrace{\frac{\Delta r_j}{r_j} + \frac{\Delta G}{G}}_{\text{aggregate output}} \right] q_{ij} + \delta \quad (3)$$

7 Decomposition of Results

This section reports the main decomposition results, decomposing emissions changes into the three channels and discusses the mechanisms underlying each component. Somewhat surprisingly, the decomposition analysis indicates that tariffs affect carbon emissions primarily through the emissions intensity channel.

Main Result

Global emissions decline by 236 million tons of CO_2 under the tariff scenario, representing less than 1 percent of projected total emissions in 2050 (30,149 million tons of CO_2). While modest in absolute terms, this reduction is still environmentally significant, as it is comparable to the annual emissions of a mid-sized economy. Table 2 reports the total net change in emissions for each region, along with the decomposition along the three channels. The final column shows that second-order effects from the linear decomposition are small, as the decomposed results closely match the observed change in emissions calculated directly from the model’s simulation results.

A key finding is that the emissions intensity channel emerges as the primary channel in the decomposition. Reductions in emissions intensity driven primarily by the adoption of cleaner production technologies and increased reliance on renewable energy account for most of the decline in emissions. The sectoral mix effect is negative but relatively small, indicating that the relative shares of “dirty” versus “clean” sectors change marginally under the tariff scenario. Finally, the reduction in emissions generated by changes in aggregate output is negligible. The reduction in emissions from lower output in regions that contract under the tariff, such as the United States, India, and China, is almost entirely offset by increases in emissions from regions that expand.

Table 2: Decomposition of Emissions Changes by Region (2050)

Region	Agg. Output	Sectoral Mix	Emissions Intensity	Total	Observed Emissions Change
ASEAN5	7.24	1.00	15.75	24.00	<i>24.03</i>
Advanced Asia	0.24	0.01	5.71	5.96	<i>5.95</i>
Caucasus and Central Asia	3.63	0.05	-0.06	3.61	<i>3.60</i>
Europe	2.71	-1.20	-32.74	-31.23	<i>-31.24</i>
India	-6.28	4.52	-34.70	-36.45	<i>-36.52</i>
Latin America	1.05	-3.15	-16.20	-18.30	<i>-18.27</i>
Middle East Oil Producers	6.15	0.76	10.78	17.69	<i>17.71</i>
Middle East and North Africa nec	8.07	0.06	-4.65	3.48	<i>3.45</i>
North America nec	-3.46	0.07	-14.78	-18.17	<i>-18.01</i>
Oceania and Pacific	0.70	0.49	6.32	7.51	<i>7.51</i>
PR China	-2.99	4.86	-98.91	-97.04	<i>-97.10</i>
Rest of World	3.14	0.22	12.97	16.33	<i>16.37</i>
Russia	5.31	-2.27	-7.21	-4.18	<i>-4.21</i>
South Asia nec	1.79	-0.03	-6.04	-4.27	<i>-4.31</i>
Southeast Asia nec	-2.61	1.07	-3.59	-5.13	<i>-5.12</i>
Sub-Saharan Africa	1.57	0.63	-4.98	-2.78	<i>-2.79</i>
United States	-27.26	-13.84	-56.43	-97.53	<i>-96.36</i>
Total	-0.99	-6.73	-228.77	-236.49	<i>-235.30</i>

Note:

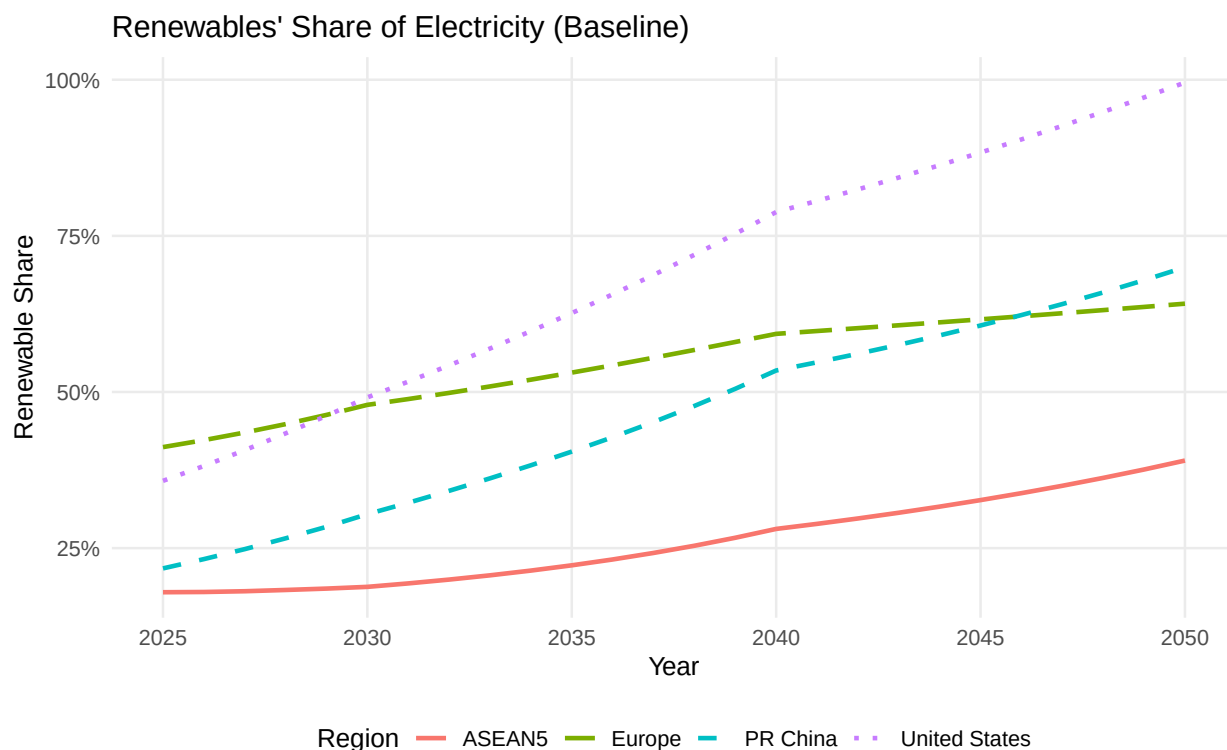
Emission changes are in million tons of CO_2

Emissions Intensity Channel

Mechanism 1: Tariffs increase the share of renewable energy

Tariffs lower the relative price of renewable energy sources by raising the cost of fossil fuels, as sizable tariffs are imposed on coal, oil, gas, and oil products, whereas electricity generation is largely exempt because it is non-tradable across borders. Through this mechanism, tariffs inadvertently increase the share of renewable energy in certain regions. It is important, however, to note that tariffs on capital equipment for renewables may still have long-term effects. Figure 11 illustrates the evolution of the renewable energy share (wind, hydro, and solar) in electricity generation over time under different green energy policies in the STEPS scenario. The green energy policy in the United States is particularly ambitious, with the share of renewables rising rapidly and approaching full decarbonization by 2050. China follows closely behind, surpassing Europe around 2045 in terms of renewable adoption. Other regions, such as the ASEAN5, also experience increases in renewable energy shares, though at a much slower pace. These gains for the ASEAN5 are even more modest under the tariff scenario, where growth in its economy and sectors dampens the expansion of clean energy. However, for regions that contract under the tariff scenario, such as the United States, Europe, and China, the impact of their green energy policies is magnified. Reduced output lowers energy demand while renewable subsidies remain fixed, resulting in a higher share of renewables in electricity generation.

Figure 11: Share of renewable energy in electricity generation under the baseline scenario



The largest declines in carbon emissions through the emissions intensity channel occur in the United States, China, Europe, and India (Table 2). The effect is especially pronounced in China, Europe, and India, which together constitute the world’s three largest emitters. Consequently, improvements in the cleanliness of production in these regions have a correspondingly larger impact on global emissions. In all three regions, the share of renewable energy in total electricity generation is higher under the tariff scenario than in the baseline (Table 3). This occurs because tariffs reduce overall electricity demand by lowering production, while subsidies and policy incentives for renewables remain fixed. With lower total demand, the unchanged environmental policies increase the renewable share of the energy mix, resulting in less emission-intensive production. The same applies to emissions from consumption, though not shown in this table, where renewable shares also rise under the tariff scenario. By contrast, regions experiencing rising emissions through this channel increase their reliance on more carbon-intensive, or “brown,” energy sources. In these cases, additional output is sustained through an expansion of fossil fuel use, which raises emissions intensity and results in dirtier production.

Table 3: Renewable Energy Share of Electric Power in 2050

Region	Renewable Share (Baseline)	Renewable Share (Policy)	% Change
ASEAN5	0.390	0.387	-0.73
Advanced Asia	0.488	0.486	-0.34
Caucasus and Central Asia	0.209	0.210	0.19
Europe	0.641	0.646	0.65
India	0.666	0.670	0.59
Latin America	0.583	0.586	0.63
Middle East Oil Producers	0.286	0.283	-0.97
Middle East and North Africa nec	0.400	0.402	0.51
North America nec	0.514	0.524	2.07
Oceania and Pacific	0.309	0.304	-1.78
PR China	0.700	0.704	0.68
Rest of World	0.326	0.323	-0.84
Russia	0.295	0.296	0.52
South Asia nec	0.430	0.438	1.78
Southeast Asia nec	0.611	0.612	0.08
Sub-Saharan Africa	0.630	0.631	0.10
United States	0.995	0.993	-0.20

Note: For the United States, the decline in the renewable share of electricity generation does not reflect a meaningful shift in energy use, as the share is already close to one in the baseline.

Mechanism 2: Other endowments are substituted for energy, making production relatively less energy-intensive

Another mechanism through which tariffs affect emissions is that they reduce the relative cost of non-energy factors compared to energy inputs. As a result, firms substitute toward non-energy

inputs, reducing energy intensity and emissions. Table 4 reports the percent changes in factor-to-energy price ratios under the tariff scenario relative to the baseline. A negative entry indicates that a given factor has become cheaper relative to energy. The results show a consistent pattern across all sectors: both skilled and unskilled labor become cheaper relative to energy. This implies that tariffs create excess labor supply in these sectors, as the demand for labor falls more sharply than the demand for energy. One reason is that labor is far more mobile across sectors than energy inputs and can therefore be reallocated more easily. Producers, therefore, substitute away from energy and toward labor, making production less energy-intensive. As production becomes less energy-intensive overall, emissions decline.

Table 4: Percent Change in Factor/Energy Price Ratios by Sector (United States, 2050)

Activity	UnSkLab	SkLab	Capital
Air_tp	-3.41	-3.49	-1.84
Chemicals	-3.09	-3.16	-1.51
Construction	-2.94	-3.01	-1.36
En_int_mfg	-1.43	-1.51	0.17
Extraction	-1.69	-1.76	-0.09
Light_mfg	-1.32	-1.40	0.28
Oth_tp	-3.27	-3.35	-1.70
Services	-1.29	-1.37	0.32
Wat_tp	-3.43	-3.51	-1.86

While labor prices fall relative to energy across the board, the adjustment is less uniform for capital. In most sectors, capital also becomes cheaper relative to energy. However, exceptions emerge in energy-intensive manufacturing, light manufacturing, and services, where capital rents remain relatively high compared to energy prices. This reflects the fact that these sectors are relatively capital-intensive, so the demand for capital is more resilient than in other sectors. Nevertheless, in the United States, most sectors substitute labor and, to some extent, capital for energy, and through this mechanism production becomes “greener.”

Sectoral Mix Channel

Tariffs shift the sectoral composition of output across regions, affecting emissions outcomes depending on which sectors expand or contract. This section focuses on sectoral changes in the United States and China and how these changes contribute to emissions outcomes. In the United States, the sectoral mix channel contributes to a reduction in overall emissions, even though energy-intensive manufacturing and light manufacturing expand significantly under the tariff scenario, by 9 percent and 6 percent respectively (Table 5). The key reason is that by 2050 manufacturing sectors in the United States are already relatively clean due to their high usage of renewable energy sources (Figure 11). Thus, contractions in high-emission sectors such as transport, consumption, and oil products, though smaller in percentage terms, generate reductions in emissions large enough

to offset the increases from expanding sectors.

Table 5: Largest Changes in Sectoral Emissions (United States, 2050)

Sector	Sector Emissions	Sector Output Chg (%)	Emission Chg
Air_tp	319.28	-2.43	-7.77
Consumption	893.08	-0.38	-3.37
Oil_Pcts	70.30	-3.86	-2.71
Wat_tp	56.08	-2.14	-1.20
Services	132.70	-0.63	-0.84
En_int_mfg	12.19	8.92	1.09
Light_mfg	7.72	5.95	0.46
Chemicals	16.60	2.66	0.44
Oth_tp	654.92	0.04	0.25
Extraction	6.99	2.64	0.18

Note:

Emissions is in million tons of CO₂.

In contrast, results from China reveal the opposite pattern, where expansion in emission-intensive sectors outweighs the reductions from contracting sectors. Under the tariff scenario, energy-intensive manufacturing expands, but unlike in the United States, this sector in China remains highly emission-intensive. Moreover, although China experiences contractions in emission-intensive sectors such as oil products and consumption, the extent of expansion in other sectors is larger, resulting in a net increase in emissions from the reallocation of sectoral production. Although production in the United States is more costly, it is considerably cleaner than in many trading partners. As a result, relocating energy-intensive industries, light manufacturing, and chemicals to the United States is environmentally beneficial. This cleaner production structure generates a net reduction of 9.6 million tons of CO₂ through the sectoral mix channel.

Table 6: Largest Changes in Sectoral Emissions (PR China, 2050)

Sector	Sector Emissions	Sector Output Chg (%)	Emission Chg
Oil_Pcts	401.99	-0.29	-1.15
Consumption	1966.95	-0.06	-1.13
Light_mfg	112.73	-0.59	-0.66
CoalE	21.41	-1.39	-0.30
Coal	41.38	-0.62	-0.26
En_int_mfg	1178.95	0.22	2.61
Air_tp	223.40	1.04	2.33
Chemicals	406.76	0.25	1.03
Services	779.56	0.13	1.00
Construction	192.98	0.29	0.55

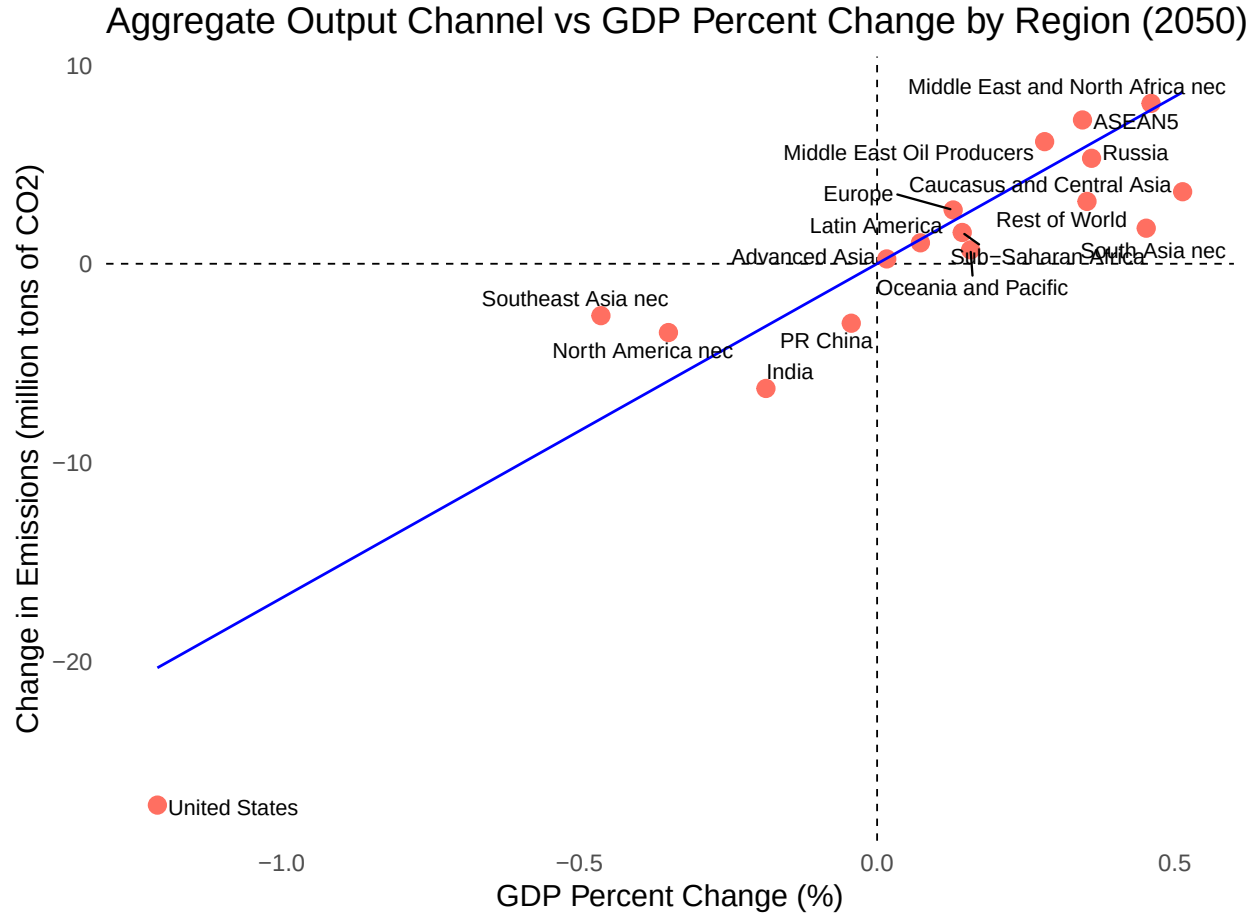
Note:

Emissions is in million tons of CO₂.

Aggregate Output Channel

Tariffs contract the global economy, leading to reduced emissions. However, some economies that are not subject to tariffs expand, resulting in increased emissions. On net, global emissions decline, but much of the reduction is offset by higher emissions from expanding economies. Figure 12 plots changes in GDP and emissions solely through the aggregate output channel. The results show that a decline in GDP leads to lower emissions, though the strength of this relationship varies across regions, reflecting differences in emissions intensities and production structures. For example, China, North America (nec), and Southeast Asia (nec) display reductions in emissions of similar magnitude, yet China achieves this outcome with a comparatively smaller contraction in GDP. This reflects the fact that Chinese production remains relatively emission-intensive, so even modest output declines translate into substantial emission reductions. By contrast, regions with cleaner production technologies require larger GDP reductions to generate comparable declines in emissions. The largest change is observed in the United States, which experiences by far the largest contraction in GDP and, correspondingly, the largest reduction in emissions. This underscores the extent to which tariffs impose economic costs on the United States and its associated emission reductions from lower economic activity.

Figure 12: Relationship between GDP and emissions changes (scale effect, including regional composition) by region in 2050 under tariff scenario.



8 Conclusion

The decomposition of emissions changes shows that the emissions intensity channel is the dominant channel for changing emissions. Two distinct mechanisms drive this result. In many regions, most notably China, Europe, and India, production becomes cleaner as the renewable share of electricity generation increases. Tariffs indirectly reinforce existing green energy policies by lowering overall electricity demand through reduced production. With environmental policies such as carbon taxes and subsidies held fixed, this reduction in electricity demand raises the renewable share of the power mix and reduces emissions intensity. A second mechanism operates through factor substitution. As tariffs reduce demand for imported intermediates and capital-intensive inputs, labor and capital become relatively cheaper than energy. This relative price shift encourages producers to substitute away from energy and toward other endowments, further lowering emissions.

The sectoral mix channel also contributes to the reduction in emissions, although the effect is small in magnitude. A key driver is the United States, where energy-intensive manufacturing

processes are reshored. Because production in these sectors is considerably cleaner in the United States than in other regions, this relocation reduces global emissions, particularly as other high-emitting U.S. sectors shrink in size. This effect outweighs the opposite tendency in other regions, where expansion is concentrated in more carbon-intensive sectors. By contrast, the aggregate output channel contributes little to overall emissions changes. Although lower GDP in some regions reduces emissions, these reductions are almost entirely offset by increases elsewhere. Consequently, the global impact of the aggregate output channel is negligible compared with the sectoral mix channel and the emissions intensity channel.

References

- Aguiar, Angel, Erwin L Corong, Dominique Van der Mensbrugghe, Eddy Bekkers, Robert Bernard Koopman, and Robert Teh**, “The WTO Global Trade Model: Technical documentation,” Technical Report, WTO staff working paper 2019.
- Ali, Saleem H.**, “Could Tariffs Improve Environmental Sustainability Outcomes?,” *Forbes*, contributor post April 2025. (accessed 5 September 2025).
- Amiti, Mary, Stephen J. Redding, and David E. Weinstein**, “The Impact of the 2018 Tariffs on Prices and Welfare,” *Journal of Economic Perspectives*, November 2019, 33 (4), 187–210.
- Antweiler, Werner, Brian R. Copeland, and M. Scott Taylor**, “Is Free Trade Good for the Environment?,” *American Economic Review*, September 2001, 91 (4), 877–908.
- Autor, David, Anne Beck, David Dorn, and Gordon H Hanson**, “Help for the Heartland? The Employment and Electoral Effects of the Trump Tariffs in the United States,” Working Paper 32082, National Bureau of Economic Research January 2024.
- Bacchetta, Marc, Eddy Bekkers, Jean-Marc Solleder, and Enxhi Tresa**, “The potential impact of environmental goods trade liberalization on trade and emissions,” *Energy Economics*, 2025, 141, 108051.
- Baldwin, Richard**, “Trumpian Tariffs Are Import Substitution Industrialisation 2.0,” 2025. VoxEU Column, Centre for Economic Policy Research (CEPR). Published online 28 August 2025. Available at: <https://cepr.org/voxeu/columns/trumpian-tariffs-are-import-substitution-industrialisation-20> (accessed 29 August 2025).
- Bekkers, Eddy and Gianmarco Cariola**, “Comparing Different Approaches to Tackle the Challenges of Global Carbon Pricing,” Technical Report ERSD-2022-10, World Trade Organization, Economic Research and Statistics Division June 2022.
- Boller, Lysle, Kody Carmody, Jon Huntley, and Felix Reichling**, “The Economic Effects of President Trump’s Tariffs,” 2025. Brief, Penn Wharton Budget Model (University of Pennsylvania). Published online 10 April 2025. Available at: <https://budgetmodel.wharton.upenn.edu/issues/2025/4/10/economic-effects-of-president-trumps-tariffs>.
- Bown, Chad P.**, “Trump’s Trade War Timeline 2.0: An Up-to-Date Guide,” 2025. PIIE RealTime Economics blog post, July 22, 2025 (updated regularly). Available at <https://www.piie.com/blogs/realtime-economics/2025/trumps-trade-war-timeline-20-date-guide> (accessed July 25, 2025).
- Brief, Carbon**, “Experts: What Do Trump’s Tariffs Mean for Global Climate Action?,” Webpage, Carbon Brief March 2025. (accessed 5 September 2025).

- Burfisher, Mary E.**, *Introduction to Computable General Equilibrium Models*, 3 ed., Cambridge University Press, 2021.
- Cherniwchan, Jevan**, “Trade liberalization and the environment: Evidence from NAFTA and U.S. manufacturing,” *Journal of International Economics*, 2017, 105, 130–149.
- , **Brian R Copeland**, and **M Scott Taylor**, “Trade and the environment: New methods, measurements, and results,” *Annual Review of Economics*, 2017, 9 (1), 59–85.
- Clausing, Kimberly**, “The Aftermath of Tariffs,” 2025. VoxEU Column, Centre for Economic Policy Research (CEPR). Published online, 29 August 2025. Available at: <https://cepr.org/voxeu/columns/aftermath-tariffs> (accessed 28 August 2025).
- Clausing, Kimberly A. and Mary E. Lovely**, “Why Trump’s Tariff Proposals Would Harm Working Americans,” 2024. Peterson Institute for International Economics Policy Brief 24-1. Published online May 20 2024. Available at: <https://ssrn.com/abstract=4834397> (accessed 29 August 2025).
- Copeland, Brian R. and M. Scott Taylor**, “North-South Trade and the Environment,” *The Quarterly Journal of Economics*, 1994, 109 (3), 755–787.
- , **Joseph S. Shapiro**, and **M. Scott Taylor**, “Globalization and the environment,” in Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff, eds., *Handbook of International Economics: International Trade*, Vol. 5, Elsevier, 2022, pp. 61–146.
- Dang, Hai-Anh H., Minh Cong Nguyen, and Trong-Anh Trinh**, “Does Hotter Temperature Increase Poverty and Inequality? Global Evidence from Subnational Data Analysis,” Policy Research Working Paper 10466, World Bank October 2023. (accessed 6 September 2025).
- Davidson, Michael R., Valerie J. Karplus, Joanna I. Lewis, Jonas Nahm, and Alex Wang**, “Risks of decoupling from China on low-carbon technologies,” *Science*, September 2022, 377 (6612), 1266–1269.
- Dean, Judith M.**, “Trade and the Environment: A Survey of the Literature,” Policy Research Working Paper 966, World Bank 1992. Background paper prepared for the World Development Report 1992. Available at: <https://econpapers.repec.org/RePEc:wbk:wbrwps:966> (accessed 31 August 2025).
- , “Does Trade Liberalization Harm the Environment? A New Test,” *Canadian Journal of Economics*, 2002, 35 (4), 819–842. (accessed 31 August 2025).
- and **Mary E. Lovely**, “Trade Growth, Production Fragmentation, and China’s Environment,” in Robert C. Feenstra and Shang-Jin Wei, eds., *China’s Growing Role in World Trade*, Chicago, IL: University of Chicago Press, 2010, pp. 429–469. Originally published as NBER chapter c10469.

- **and** –, “Global Value Chains, Foreign Direct Investment, and Industrial Pollution Intensity in China,” *Review of International Economics*, 2025, *33* (3), 741–764.
- , – , **and Hua Wang**, “Are foreign investors attracted to weak environmental regulations? Evaluating the evidence from China,” *Journal of Development Economics*, 2009, *90* (1), 1–13.
- Evenett, Simon and Johannes Fritz**, “US Reciprocal Tariffs: Upending the Global Trade Policy Landscape,” 2025. CEPR VoxEU Column, April 3, 2025. Available at <https://cepr.org/voxeu/columns/us-reciprocal-tariffs-upending-global-trade-policy-landscape>.
- Fajgelbaum, Pablo D., Pinelopi K. Goldberg, Patrick J. Kennedy, and Amit K. Khandelwal**, “The Return to Protectionism,” *The Quarterly Journal of Economics*, 2019, *135* (1), 1–55. .eprint: <https://academic.oup.com/qje/article-pdf/135/1/1/32526166/qjz036.pdf>.
- Grantham-Philips, Wyatte**, “Trump’s Tariffs Have Launched Global Trade Wars. Here’s a Timeline of How We Got Here,” 2025. Associated Press, May 12, 2025. Available at <https://apnews.com/article/trump-tariff-england-trade-war-e8300b457256d4c23d64928c09ce2b63> (accessed July 25, 2025).
- Grossman, Gene M. and Alan B. Krueger**, “Environmental Impacts of a North American Free Trade Agreement,” in Peter M. Gerber, ed., *The U.S.-Mexico Free Trade Agreement*, Cambridge, MA: MIT Press, 1993, pp. 13–56.
- Guo, Jessie, Daniel Kubli, and Patrick Saner**, “The Economics of Climate Change: No Action Not an Option,” Expertise Publication, Swiss Re Institute April 2021. (accessed 5 September 2025).
- Hu, Xiurong, Hector Pollitt, Jamie Pirie, Jean-Francois Mercure, Junfeng Liu, Jing Meng, and Shu Tao**, “The impacts of the trade liberalization of environmental goods on power system and CO2 emissions,” *Energy Policy*, 2020, *140*, 111173.
- IEA**, “Net Zero by 2050: A Roadmap for the Global Energy Sector,” Special Report, International Energy Agency May 2021. Revised version, October 2021. Licence: CC BY 4.0. Available as PDF: https://iea.blob.core.windows.net/assets/deebef5d-0c34-4539-9d0c-10b13d840027/NetZeroBy2050-ARoadmapfortheGlobalEnergySector_CORR.pdf (accessed 6 September 2025).
- IEA**, “Global Energy and Climate Model Documentation,” Technical Report, International Energy Agency October 2024. Revised version, October 2024; CC BY 4.0 licence.
- J.P. Morgan Global Research**, “US tariffs: What’s the impact on global trade and the economy?,” 2025. J.P. Morgan Global Research, published online 11 July 2025. Available at: <https://www.jpmorgan.com/insights/global-research/current-events/us-tariffs> (accessed 29 August 2025).

- Kotz, Maximilian, Anders Levermann, and Leonie Wenz**, “The Economic Commitment of Climate Change,” *Nature*, 2024, 628 (8008), 551–557. (accessed 6 September 2025).
- Levinson, Arik**, “Are Developed Countries Outsourcing Pollution?,” *Journal of Economic Perspectives*, September 2023, 37 (3), 87–110.
- Li, Taipeng, Lorenzo Trimarchi, Rui Xie, and Guohao Yang**, “The unintended consequences of trade protection on the environment,” *Journal of Development Economics*, 2026, 178, 103547.
- Liu, Li-Jing, Felix Creutzig, Yun-Fei Yao, Yi-Ming Wei, and Qiao-Mei Liang**, “Environmental and economic impacts of trade barriers: The example of China–US trade friction,” *Resource and Energy Economics*, 2020, 59, 101144.
- Lowell, Michael, Philippe Heeren, Justin Angotti, Lizbeth Rodriguez-Johnson, Kirsten Lowell, and Courtney E. Fisher**, “Trump 2.0 Tariff Tracker,” 2025. Trade Compliance Resource Hub (Reed Smith LLP). Published online August 29, 2025. Available at: <https://www.tradecomplianceresourcehub.com/2025/08/29/trump-2-0-tariff-tracker/#updates> (accessed 31 August 2025).
- Mahoney, Neale and Ryan Cummings**, “Framing the Next Four Years: Tariffs, Tax Cuts and Other Uncertainties in the Trump Administration,” 2025. Policy Brief, Stanford Institute for Economic Policy Research (SIEPR). Published online January 2025. Available at: <https://siepr.stanford.edu/publications/policy-brief/framing-next-four-years-tariffs-tax-cuts-and-other-uncertainties-trump> (accessed 29 August 2025).
- McKibbin, Warwick J., Marcus Noland, and Geoffrey Shuetrim**, “The Global Economic Effects of Trump’s 2025 Tariffs,” 2025. Working Paper 25-13, Peterson Institute for International Economics (PIIE). Published online June 2025. Available at: <https://www.piie.com/publications/working-papers/2025/global-economic-effects-trumps-2025-tariffs> (accessed 29 August 2025).
- Obstfeld, Maurice**, “Trump’s Tariffs Are Designed for Maximum Damage to America,” 2025. RealTime Economics Blog, Peterson Institute for International Economics, Washington, DC, April 4, 2025. Available at: <https://www.piie.com/blogs/realtime-economics/2025/trumps-tariffs-are-designed-maximum-damage-america> (accessed January 15, 2026).
- Office of the United States Trade Representative**, “Presidential Tariff Actions,” 2025. Accessed July 25, 2025. Available at <https://ustr.gov/trade-topics/presidential-tariff-actions>.
- Petri, Peter A., Michael G. Plummer, Cyn-Young Park, and Sean Hwang**, “The Economics of the Net Zero Transition: Policy Scenarios and the Role of Trade and Cooperation,”

- <https://ssrn.com/abstract=5448397> or <http://dx.doi.org/10.2139/ssrn.5448397> 2025. SSRN Working Paper, August 29, 2025.
- PIIE**, “PIIE Experts React to Trump’s Tariffs Announced April 2,” 2025. RealTime Economics Blog, Peterson Institute for International Economics, April 2, 2025. Available at: <https://www.piie.com/blogs/realtime-economics/2025/piie-experts-react-trumps-tariffs-announced-april-2> (accessed January 15, 2026).
- Sekin, Figen**, “US Tariffs and The Damaging Impact on Sustainability in 2025,” WinS Solutions Resource Hub, website article April 2025. (accessed 5 September 2025).
- Storrow, Benjamin, Sara Schonhardt, Mike Lee, and E&E News**, “Trump’s Tariffs Are Expected to Undermine the Clean Energy Transition,” Webpage, Scientific American (Climatewire) April 2025. (accessed 5 September 2025).
- World Bank**, “Global Action Urgently Needed to Halt Historic Threats to Poverty Reduction,” Feature Story, World Bank Website October 2020. (accessed 6 September 2025).
- , *World Development Report 2020: Trading for Development in the Age of Global Value Chains*, Washington, DC: World Bank, 2020.
- World Trade Organization**, “World Trade Report 2022: Climate Change and International Trade,” WTO World Trade Report, World Trade Organization June 2022. (accessed 6 September 2025).
- , “Trade Policy Tools for Climate Action,” Policy Toolkit / Report, World Trade Organization December 2023. (accessed 6 September 2025).
- York, Erica and Alex Durante**, “Trump Tariffs: Tracking the Economic Impact of the Trump Trade War,” 2025. Tax Foundation, published online 5 May 2025. Available at: <https://taxfoundation.org/research/all/federal/trump-tariffs-trade-war/> (accessed 29 August 2025).
- Zhong, Jiarui and Jiansuo Pei**, “Carbon border adjustment mechanism: a systematic literature review of the latest developments,” *Climate Policy*, 2024, 24 (2), 228–242.

Appendix

Table 7: Region Code Mapping

#	Code	Region Name
1	chn	PR China
2	as5	ASEAN5
3	sel	Southeast Asia nec
4	ind	India
5	xsa	South Asia nec
6	cca	Caucasus and Central Asia
7	dea	Advanced Asia
8	ocn	Oceania and Pacific
9	usa	United States
10	xna	North America nec
11	lac	Latin America and Caribbean
12	eur	Europe
13	meo	Middle East Oil Producers
14	mna	Middle East and North Africa nec
15	afr	Sub-Saharan Africa
16	rus	Russia
17	row	Rest of World

Table 8: Sector List with Descriptions

#	Sector Code	Description
1	Crops	Agricultural crops
2	Livestock	Animal husbandry and meat production
3	Extraction	Mining and raw resource extraction
4	Chemicals	Chemical and petrochemical products
5	Light_mfg	Light manufacturing (e.g., textiles, apparel)
6	En_int_mfg	Energy-intensive manufacturing (e.g., metals, cement)
7	Electrical	Electrical equipment manufacturing
8	Machinery	Industrial and general-purpose machinery
9	Trans_eq	Transport equipment (e.g., vehicles, aircraft)
10	TnD	Transmission and distribution (electricity grid)
11	Construction	Construction services
12	Dwellings	Residential housing services
13	Services	Other commercial and public services
14	Oth_tp	Other transport (unspecified)
15	Wat_tp	Water transport
16	Air_tp	Air transport
17	Coal	Coal mining and processing
18	Oil	Crude oil extraction
19	Gas	Natural gas extraction
20	Oil_Pcts	Refined petroleum products
21	NuclearE	Nuclear power generation
22	CoalE	Coal-based electricity generation
23	GasE	Gas-based electricity generation
24	WindE	Wind power generation
25	HydroE	Hydroelectric generation
26	OilE	Oil-based electricity generation
27	OthE	Other renewable or miscellaneous energy sources
28	SolarE	Solar power generation

Emission Shares by Region (Top 5 + Rest of the World, Baseline)

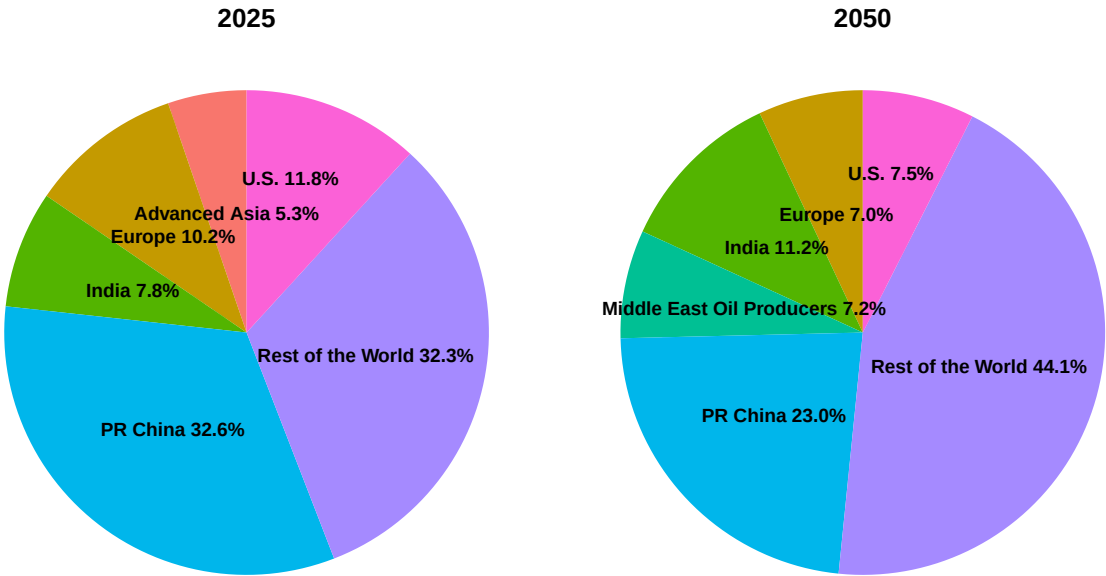


Figure 13: Percent change in emissions by country and region in 2050 under tariff scenario (relative to baseline).

Emissions by Region in 2050: Base vs. Policy

